Assignment Part B – Overview

Wiki Content Management and Admin Console

Extend your Part 1 login system into a Spring Boot MVC wiki-style web application using Thymeleaf. The application should allow public viewing of articles and provide an admin area for managing articles and categories.

The task focuses on understanding MVC flow and layered design using Controllers, Services, Repositories, and Thymeleaf views.

Remember too that a smaller, working application is better than a large, unfinished application.

[Click on triangles to expand headings]

Requirements

See the 2nd Part B handout.

Demonstration (compulsory)

You must attend an in-class demonstration in Week 13 or 14.

During the demonstration, you may be asked to:

- Explain how a request moves from the browser to your Controller
- Show how your Controller sends data to the view
- Describe what your Service class does
- Explain how your Repository stores or retrieves data
- Describe how your model represents your content
- Explain how your CRUD workflow works
- Explain how admin access is restricted
- Modify or explain a small section of your code

You are not required to explain every line of code, but you must demonstrate clear understanding of the overall system and the key components. If you cannot explain your implementation, your marks may be reduced even if the system appears to work.

If you can clearly explain "I click this → it goes to this Controller → calls this Service → uses this Repository → changes or retrieves the data → returns to this page", then you are meeting the core objective of this assessment.

Submission requirements

Due date

See Moodle submission/upload folder for submission date.

How

Submit **one ZIP file** to Moodle containing:

- Your complete Spring Boot project
- All source code
- Templates
- Static resources
- Database-related files if required

Do **not** submit the **target** folder or other unnecessary generated files.

Regarding **database**-related files:

- Your application should run without requiring major reconfiguration by the lecturer.
- If you are using H2, ensure your application can run directly after extraction. Also, include any required H2 database files if your project depends on saved data
- If you are using MySQL Workbench or similar:
    - Include an SQL export file containing the database structure and data
    - Include clear instructions describing:
        - Database name
        - Username/password if required
        - Any setup steps needed before running the application

Marking Scheme

This assessment includes an in-class individual demonstration that supports authentic authorship and academic integrity.

Demonstration Component (SLOs 1, 2, 3, & 4)

Students will first receive a technical mark (out of 35) based on the marking rubric below. The final percentage grade for this assessment will then be adjusted according to demonstration performance as follows:

Demonstration Grade	Adjustment
HD (85–100%)	No adjustment
D (75–84%)	Final mark = 95% of technical mark
C (65–74%)	Final mark = 85% of technical mark
P (50–64%)	Final mark = 70% of technical mark
N (0–49%)	Final mark capped at 49%

Students who are unable to explain their own architecture, MVC flow, database integration, authentication logic, or key application behaviour will receive a demonstration grade below 50%, and their overall Part B mark will not exceed 49%.

Rubric (SLOs 1, 2, 3, & 4)

The rubric is used holistically. Small technical mistakes should not automatically prevent students from achieving higher grades if the overall system demonstrates strong understanding and integration.

Criteria	HD 80%-100%	D 70%-79%	C 60%-69%	P 50%-59%	N 0%-49%
Working Application and Integration 6 Marks. SLOs 1, 2, 3.	Application runs successfully and major features integrate correctly. Navigation, login, CRUD, and page flow operate reliably.	Application works with only minor integration issues. Most features function correctly.	Core functionality works but some integration or navigation issues exist.	Application partially functions but contains major integration problems.	Application does not run correctly, cannot be demonstrated successfully, or major functionality is missing.
Interface and Navigation 4 Marks. SLOs 1, 3.	Interface is clear, readable and well organised. Navigation is consistent across pages. CSS and static resources used appropriately.	Interface clear and functional with minor inconsistencies.	Interface generally usable but layout or navigation issues evident.	Interface difficult to use or inconsistent in several places.	Interface incomplete or largely unusable.
MVC Structure and Code Organisation 5 Marks. SLOs 1, 2, 3.	Strong MVC layering with correct Controller-Service-Repository separation. File structure and code organisation follow Spring Boot conventions.	Layering mostly correct with only minor structural issues.	Basic MVC structure present but inconsistently applied.	Weak separation of concerns or inconsistent organisation.	MVC structure missing or poorly implemented.
Authentication and Session Management 5 Marks. SLOs 1, 2, 3.	Login system works correctly with hashed passwords, restricted admin access, and reliable session management.	Authentication mostly correct with minor flaws.	Basic authentication and session handling implemented with some issues.	Authentication partially implemented or inconsistent.	Authentication missing or non-functional.

Criteria	HD 80%-100%	D 70%-79%	C 60%-69%	P 50%-59%	N 0%-49%
CRUD Functionality 7 Marks. SLOs 1, 2, 3.	CRUD operations implemented correctly and reliably. Data updates and retrieval function consistently.	Most CRUD operations work correctly with minor issues.	Core CRUD functionality present but contains logical or structural problems.	Partial CRUD implementation only.	CRUD functionality missing or largely non-functional.
Database and Entity Integration 3 Marks. SLOs 2, 3.	Models, relationships, and persistence implemented correctly. Database integration reliable and well structured.	Database integration mostly correct with minor issues.	Basic persistence and entity structure implemented with inconsistencies.	Limited or partially functioning database integration.	Database integration missing or incorrect.
Validation and Reliability 2 Marks. SLO 3.	HTML and JavaScript validation implemented appropriately. Invalid input handled sensibly.	Validation implemented with minor gaps.	Basic validation present but limited in effectiveness.	Minimal validation implemented.	Validation missing.
JUnit Testing 2 Marks. SLO 3.	Meaningful JUnit tests implemented and demonstrate understanding of testing and validation.	JUnit tests implemented with minor gaps.	Basic tests present but limited in depth or coverage.	Minimal or incomplete testing.	No meaningful testing provided.
Comments & Code Readability 1 Mark. SLOs 3, 4.	Code is readable and appropriately commented. Naming conventions clear and meaningful.	Code mostly readable with minor issues.	Basic readability demonstrated.	Readability inconsistent or comments limited.	Code difficult to follow or lacks comments.

Total: 35 Marks